mammals; reptiles and amphibians; birds; fish and aquatic animals; and snakes and scorpions.

The invention of the microscope also had a major impact. English scholar Robert Hooke was quick to adopt this new technology: his book *Micrographia* (1665) inspired others to do likewise. Able to view specimens magnified to 50 times their actual size, he made meticulous drawings of microscopic life, and also coined the term “cell” after examining plant fibers. Hooke also suggested a living origin for fossil fragments found in rocks.

Classifying variety

English vicar John Ray’s *History of Plants* (1686–1704) was the botanical equivalent of Gesner’s earlier work, listing some 18,000 species in three huge volumes. Ray also produced a biological definition of a species, remarking that “one species never springs from the seed of another.”

Swedish botanist Carl Linnaeus, the “father of taxonomy,” first published *Systema Naturae* in 1735, but it is the 10th edition (1758) that founded the modern zoological naming system. Two volumes of Linnaeus’ work are devoted to plants and animals, which he divided into classes, orders, genera, and species. The binomial system, in which every species is given a generic name followed by a specific name, is still in use today. Linnaeus also wrote a third volume on rocks, minerals, and fossils.

Species concepts

Building on Darwin’s theory of evolution by natural selection, German-American evolutionary biologist Ernst Mayr cemented the biological concept of species in his *Systematics and the Origin of Species* (1942). He argued that a species is not just a group of morphologically similar individuals, but a population that can breed only among themselves. Mayr